KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY (KUET)
Department of Computer Science and Engineering
Course No: CSE 4122 **Course Title:** Natural Language Processing Laboratory

Project Proposal: DeClickBait

Detecting and Correcting Multilingual Clickbait Headlines Using Discriminative and Generative NLP

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1 Project Motivation and Scope

Digital journalism and social platforms heavily exploit sensationalist headlines that exaggerate claims, withhold key information, or induce artificial emotional urgency to maximize clicks. In South Asian digital media, this problem is compounded by widespread linguistic code-mixing across **Standard Bengali**, **Romanized Banglish**, and **English**.

Existing solutions treat clickbait handling almost exclusively as a binary headline classification problem. While flagging deceptive content is helpful, it fails to inform the reader of what the article actually contains. **DeClickBait** reformulates this challenge into a dual-task objective: (i) reliably identifying whether a given headline is clickbait by cross-referencing the article's body text, and (ii) autonomously generating a truthful, fact-grounded replacement headline directly from the source content.

2 Architectural Strategies: Separate vs. Multi-Head Modeling

Given an input pair consisting of a candidate headline and article body text, the project explores and compares two architectural paradigms:

- **Approach 1: Decoupled Two-Model Pipeline**: An architecture employing two dedicated neural models operated sequentially. The first is a discriminative evidence classifier that analyzes semantic alignment and discrepancy between headline and body representations to output a clickbait posterior probability. If flagged as clickbait, the input is passed to an independent sequence-to-sequence summarization model that ingests the body text to decode a factual replacement headline.
- **Approach 2: Unified Multi-Head Neural Architecture**: An end-to-end framework where a single shared text encoder extracts unified contextual embeddings across languages, branching into two concurrent heads: a discriminative classification head and a generative decoding head. This approach drastically reduces overall memory footprint and shares lexical features, while requiring careful multi-task optimization to prevent generative gradients from destabilizing classification features.

3 Embedding Strategies and Linguistic Representation

Because Bangladeshi digital media frequently transitions between Bengali script and Romanized phonetics, the project investigates two primary representation strategies:

- **Pretrained Multilingual Transformers**: Leveraging large-scale domain-adapted pretrained models (such as Banglish-BERT and multilingual subword tokenizers) to transfer rich grammatical and semantic priors.
- **From-Scratch Domain-Specific Subword Embeddings**: Training specialized subword representations (e.g., Byte-Pair Encoding) directly on curated news and social media corpora to evaluate whether compact, domain-targeted embeddings can match heavy pretrained models while drastically reducing latency and memory requirements.

4 Evaluation Plan and Key Deliverables

The project will benchmark models on balanced multilingual test splits across the three language modalities (Bengali, Banglish, and English):

- **Classification Evaluation**: Measured using Precision, Recall, Macro-F1 score, and ROC-AUC under optimal decision thresholding.
- **Generation Evaluation**: Assessed via ROUGE metrics (ROUGE-1, ROUGE-2, ROUGE-L) and Perplexity against verified ground-truth headlines, supplemented by human qualitative checks for factual consistency.
- **Target Deliverables**: (i) A documented, curated multilingual clickbait dataset; (ii) comparative evaluation of separate vs. multi-head model architectures; (iii) an efficient, unified inference pipeline providing real-time clickbait detection and headline remediation; and (iv) a comprehensive research report.